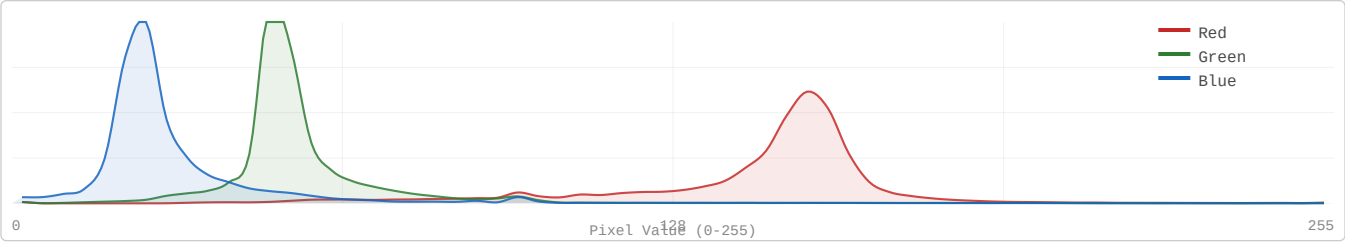
Decorrelation Gap: KODIM02

768x512 | 24-bit RGB | PNG (lossless)

Red Door Latch and Bolt | Three-dimensional (PC1 = 73.37%)

1. RGB Channel Distribution



2. Inter-Channel Correlation: RGB vs YCbCr

RGB (Original)

	R	G	B
R	**1.0000**	0.5279	0.4063
G	0.5279	**1.0000**	0.9740
B	0.4063	0.9740	**1.0000**

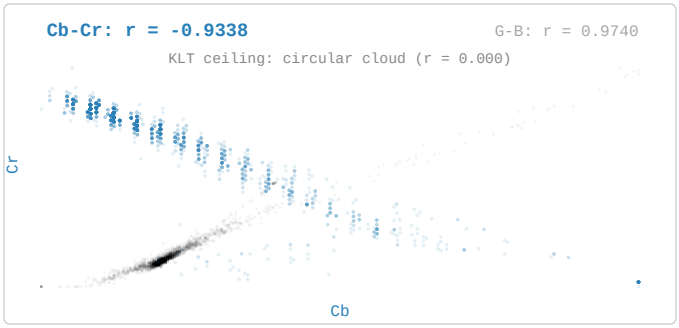
Cb-Cr: -0.934

(KLT ceiling = 0.000)

YCbCr (BT.601)

	Y	Cb	Cr
Y	**1.0000**	-0.2038	0.0591
Cb	-0.2038	**1.0000**	-0.9338
Cr	0.0591	-0.9338	**1.0000**

3. Cb vs Cr Scatter with G-B background overlay



4. Pairwise Decorrelation Gap

Channel Pair	RGB	YCbCr (BT.601)	Residual |r|
R-G → Y-Cb	0.527910	-0.203812	0.2038
R-B → Y-Cr	0.406272	0.059137	0.0591
G-B → Cb-Cr	0.973974	-0.933788	0.9338
Avg |r|	0.636052	0.398912	0.3989

Decorrelation Gap: KODIM02

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5. Decorrelation Efficiency & Blue Independence

Decorrelation Efficiency

37.3% Low

0% BT.601 leaves all — 100% KLT ceiling

Blue Channel Independence (Baetzel 2026a)

27.4%

0% Fully predictable — 100% Fully independent

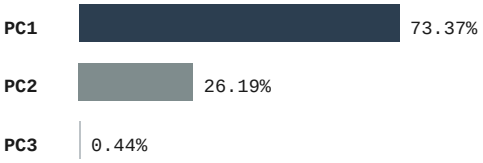
6. 4:2:0 Chroma Subsampling — Information Cost

Channel	PSNR (dB)
R	39.87 dB
G	46.86 dB
B	42.79 dB
Overall	42.31 dB

Method: BT.601 RGB→YCbCr | 2x box downsample Cb,Cr | nearest upsample | BT.601 inverse

7. PCA Baseline Reference (Baetzel 2026a)

Eigendecomposition



Eigenvector Loadings

	R	G	B	Eigenvalue
PC1	-0.6501	-0.5642	-0.5089	1234.91
PC2	-0.7550	0.4043	0.5162	440.82
PC3	0.0855	-0.7198	0.6889	7.47

8. PCA Derived Metrics

Condition Number

165.41

PC1/PC2: 2.80 | PC2/PC3: 59.05

Blue Channel Independence

27.4%

Variance not captured by PC1

PC1 Dominant Channel

R

Loading: R (0.65)

9. Redundancy Profile

Classification: Three-dimensional | CN: 165.41 | PC1: 73.4%

RGB Correlation: Avg |r| = 0.636052 | R-G: 0.5279 | R-B: 0.4063 | G-B: 0.9740

YCbCr Residual: Avg |r| = 0.398912 | Y-Cb: -0.2038 | Y-Cr: 0.0591 | Cb-Cr: -0.9338

Efficiency: 37.3% of KLT ceiling | Gap: 0.3989 avg |r| remaining

Blue Channel: 27.4% independent | 4:2:0 PSNR: 42.31 dB

References

[1] Baetzel, J. (2026a). "Statistical Characterization of Inter-Channel Redundancy Structure"
[2] Wallace, G.K. "The JPEG still picture compression standard." IEEE Trans. CE 38(1), 1992.
[3] ITU-R Recommendation BT.601: Studio Encoding Parameters, 1982.
[4] Watanabe, S. "Karhunen-Loeve Expansion and Factor Analysis," pp. 635-660, 1965.